DATA SCIENCE MINOR PROJECT

(INT 387)

PROJECT REPORT

(Project Semester January - April 2026)

NYC GRADUATION RESULT DASHBOARD

Interactive Power BI Dashboard for NYC Graduation Results Analysis

Cohorts 2012–2019 | Data.gov Dataset

Submitted by

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Course Code: INT 387

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Declaration

I hereby declare that this project report titled **"NYC Graduation Results Dashboard Analysis"** is my original work and has been completed by me as part of my academic requirements. I confirm that this project has not been submitted earlier by any student for any purpose.

All the data and information used in this project have been taken from reliable sources and have been properly analyzed using appropriate tools and techniques. I take full responsibility for the authenticity and accuracy of the work presented in this report.

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Section: 2M030
Date: 19 April 2026

Acknowledgement

I would like to extend my heartfelt thanks to my teacher **Mrs. Ashima**, who guided me and supported me throughout this project, thereby helping me learn all the necessary things to complete the report.

I also want to thank my institute for providing me with the chance to work on such a wonderful project that allowed me to gain additional knowledge and skills.

Moreover, I am thankful to my friends and relatives who provided me with constant support during the project completion.

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Lovely Professional University

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1. Introduction

This project focuses on analyzing the graduation performance of students in New York City (NYC) from 2012 to 2019 using data visualization techniques. The main objective of this study is to understand trends in graduation rates, dropout rates, and overall academic achievements. The dataset includes information from different boroughs such as Bronx, Brooklyn, Manhattan, Queens, and Staten Island, which allows a detailed comparison of student performance across regions.

A Power BI dashboard is created to present the data in a visual and interactive format using charts, graphs, and maps. These visualizations make it easier to identify patterns, track changes over time, and compare results between different boroughs. The dashboard also helps in simplifying complex data so that meaningful insights can be easily understood.

The analysis highlights that graduation rates have generally improved over the years, while dropout rates have decreased, although some regions still face challenges. Differences in performance among boroughs are clearly visible, with some areas performing better than others. Additionally, the study of Advanced Regents achievements shows trends in higher academic success. Overall, this project provides valuable insights that can help in improving educational strategies and decision-making.

31M	3M	39M
Total Graduates	Dropout Count	Total Students

2. Source of Dataset

The dataset used in this project is based on publicly available educational data of New York City (NYC) schools. It includes information about student graduation rates, dropout rates, total enrolment, and academic achievements across different boroughs such as Bronx, Brooklyn, Manhattan, Queens, and Staten Island. The data covers multiple years from 2012 to 2019, allowing analysis of trends over time.

Dataset Name	NYC Graduation Results
Source	https://catalog.data.gov/dataset/graduation-results-for-cohorts-2012-to-2019-classes-of-2016-to-2023
Cohort Years	2012 to 2019
Geographic Coverage	Bronx, Brooklyn, Manhattan, Queens, Staten Island
Key Variables	Grads %, Advanced Regents %, Dropout %, Still Enrolled, SACC (IEP/TASC)
File Format	CSV / Excel (imported into Power BI)

This dataset is typically sourced from NYC Open Data and educational reports published by the New York City Department of Education. It is widely used for research and analysis purposes to understand student performance and improve educational outcomes.

Before using the dataset for analysis, it was carefully reviewed to ensure accuracy and consistency. The data was cleaned by handling missing values and organizing it into a structured format suitable for visualization. This preparation step is important to ensure that the results obtained from the analysis are reliable and meaningful.

3. Dataset Preprocessing

Data preprocessing is an important step in this project to ensure that the dataset is clean, accurate, and ready for analysis. The raw data may contain missing values, errors, or inconsistent formats, which can affect the results if not handled properly. Therefore, necessary steps were taken to improve data quality before visualization.

3.1 Data Cleaning

Data cleaning is the process of removing errors and fixing issues in the dataset.

- Removed missing or blank values in columns like graduation rate or dropout rate
- Checked and removed duplicate records
- Corrected inconsistent column names
- Ensured values like percentages were in correct format

3.2 Data Transformation

Data transformation means converting data into a suitable format for analysis.

- Converted data types (e.g., year → numeric, percentage → decimal)
- Organized data into proper columns like: Graduation Rate, Dropout Rate, Total Students
- Filtered data for years 2012–2019 only

4. Analysis on Dataset

The dataset is analyzed to study graduation trends, dropout rates, and student performance across different boroughs of NYC. It helps in identifying patterns, comparing regions, and understanding overall student outcomes using visualizations.

Obj 1	**Graduation Trend Analysis** To analyze the trend of graduation rates from cohort years 2012 to 2019 and determine how much it has improved or declined over time.

I. General Description

This objective focuses on analyzing the trend of graduation rates over the period from 2012 to 2019. The purpose is to understand how student performance has changed over time and whether there is improvement or decline in graduation outcomes.

ii. Specific Requirements

- Cohort Year (2012–2019)
- Graduation Rate (%)
- Dropout Rate (%)
- Advanced Regents of Cohort (%)

iii. Analysis Results

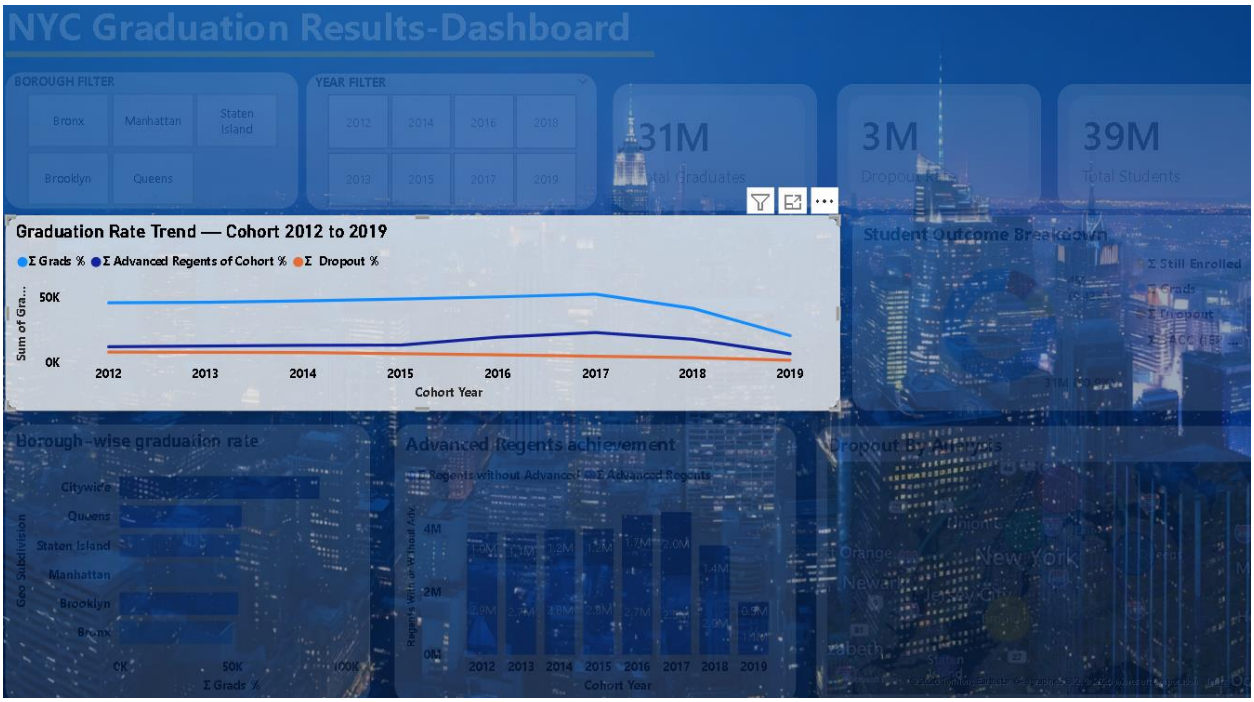
The analysis shows that the graduation rate steadily increased from 2012 to 2017, indicating improvement in the education system and student performance. However, after 2017, there is a slight decline in graduation rates, which may indicate challenges such as academic pressure or external factors. At the same time, dropout rates have generally decreased over the years, which is a positive sign as fewer students are leaving school without completing their education.

iv. Visualisation

Visualisation Type: Line Chart (Multi-series)

Visual Elements used in Power BI Dashboard:

- X-Axis: Cohort Year (2012 to 2019)
- Y-Axis: Sum of Grads % , Advanced Regents of Cohort% , Dropout %
- Chart titled "Graduation Rate Trend "



Obj 2	**Borough-wise Performance Comparison** To compare the graduation rates of Bronx, Brooklyn, Manhattan, Queens, and Staten Island and identify which borough performs the best

i. General Description

This objective provides a comparative analysis of graduation rates across the five boroughs of New York City. Each borough has unique demographic, socioeconomic, and institutional characteristics that may influence student outcomes. Understanding borough-level performance disparities is essential for targeted resource allocation and policy intervention.

ii. Specific Requirements

- Borough (Bronx, Brooklyn, Manhattan, Queens, Staten Island)
- Graduation Rate (%)
- Total Students

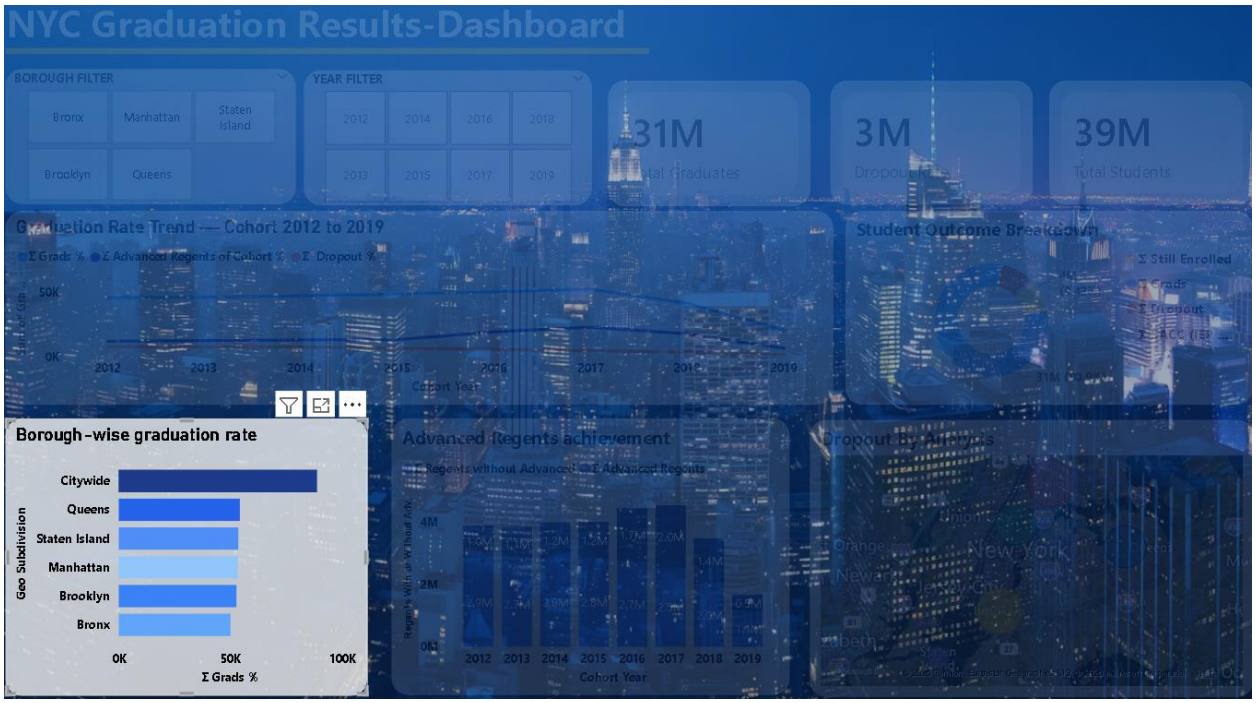
iii. Analysis Results

The analysis reveals that Queens and Staten Island have higher graduation rates, indicating better academic performance and educational support systems. On the other hand, Bronx shows lower graduation rates and relatively higher dropout rates, suggesting that it may face more educational challenges. Brooklyn and Manhattan fall in the middle range, showing moderate performance. This comparison highlights inequality in education across different regions.

iv. Visualisation

Visualisation Type: Horizontal Bar Chart

- Y-Axis (Geo Subdivision): Citywide, Queens, Staten Island, Manhattan, Brooklyn, Bronx
- X-Axis: Sum of Grads % (0K to 100K range)
- Bars displayed in uniform blue for clean comparative reading
- Chart titled "Borough-wise graduation rate"


NYC Graduation Results-Dashboard
BOROUGH FILTER
Bronx
Manhattan
Staten Island
Brooklyn
Queens
YEAR FILTER
2012
2014
2016
2018
2013
2015
2017
2019
31M
3M
39M
Total Students
Graduation Rate Trend — Cohort 2012 to 2019
Σ Grads %
Σ Advanced Regents of Cohort %
Σ Dropout %
50K
0K
2012
2013
2014
2015
2016
2017
2018
2019
Cohort Year
Student Outcome Breakdown
Σ Still Enrolled
Σ Grads
Σ Dropout
Borough-wise graduation rate
Citywide
Queens
Staten Island
Manhattan
Brooklyn
Bronx
Geo Subdivision
0K
50K
100K
Σ Grads %
Advanced Regents achievement
Regents without Advanced
Σ Advanced Regents
4M
2M
0M
Cohort Year
Dropout By Ethnicity
Orange
New York
Newark
Elizabeth

Obj 3	**Student Outcome Breakdown** To analyze the percentage of students who are Graduated, Dropout, Still Enrolled, and in IEP/TASC categories in order to understand overall student outcomes.

i. General Description

This analysis focuses on understanding student graduation performance in New York City (NYC) from 2012 to 2019. It examines key factors such as graduation rates, dropout rates, and student outcomes across different boroughs. The purpose is to identify trends over time, compare regional performance, and evaluate overall student success. By analyzing this data, it becomes easier to understand patterns of improvement or decline and highlight areas that require attention for better educational outcomes.

ii. Specific Requirements

- Number of Graduates
- Number of Dropouts
- Still Enrolled Students
- IEP/TASC StudentS.

iii. Analysis Results

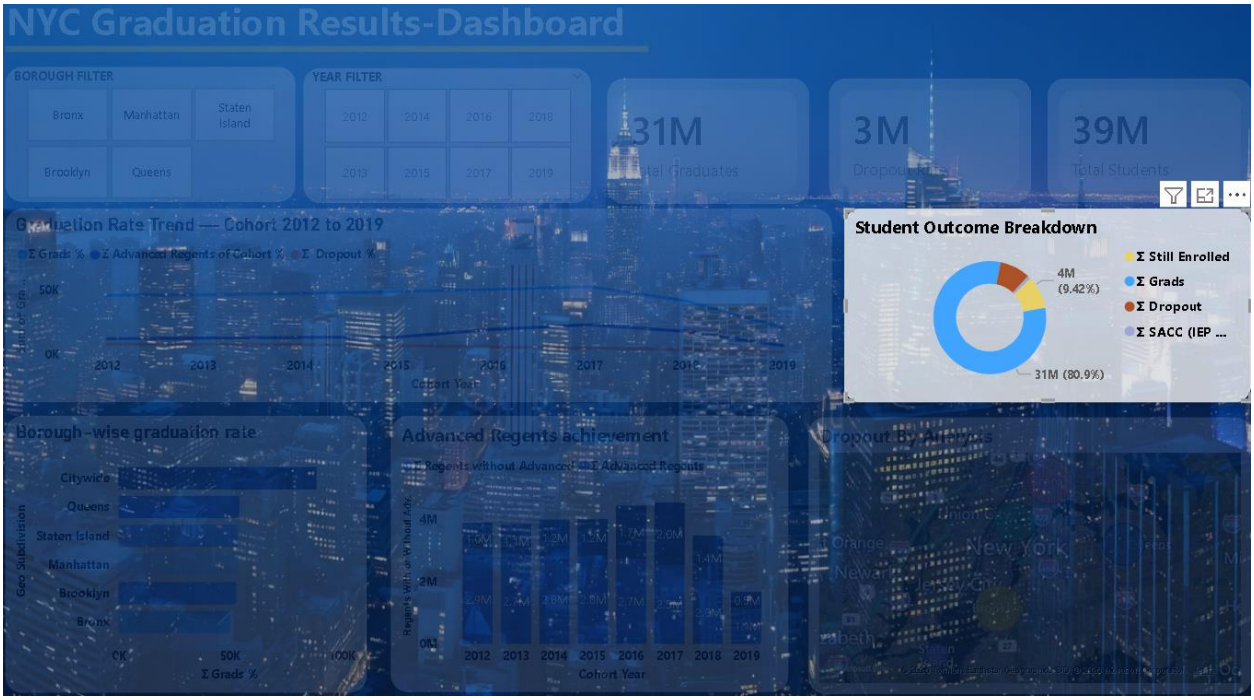
The results show that a large majority of students, approximately 80%, successfully graduate. A smaller percentage of students drop out, while some remain enrolled or fall into special categories such as IEP or TASC programs. This indicates that while overall performance is good, there is still a need to focus on reducing dropout rates and supporting students who are not completing their education.

iv. Visualisation

Visualisation Type: Donut Chart

- Four colour-coded segments: Yellow (Still Enrolled), Blue (Grads), Orange (Dropout), Purple (SACC/IEP)
- Percentage labels displayed for each segment

- Central annotation showing dominant value (31M — 80.9%)
- Legend positioned to the right for clear segment identification



Obj 4	**Geographic Dropout Risk Analysis** To analyze dropout rates across the five boroughs of NYC using a map visualization, and identify high-risk and low-risk areas through bubble size and color.

i. General Description

This analysis examines dropout rates across different geographic locations to identify high-risk and low-risk areas. It helps in understanding which regions have higher dropout issues and require more attention.

ii. Specific Requirements

- Geographic mapping of all five NYC boroughs: Bronx, Brooklyn, Manhattan, Queens, Staten Island
- Borough Location
- Dropout Rate (%)
- Geographic Coordinates / Map Data

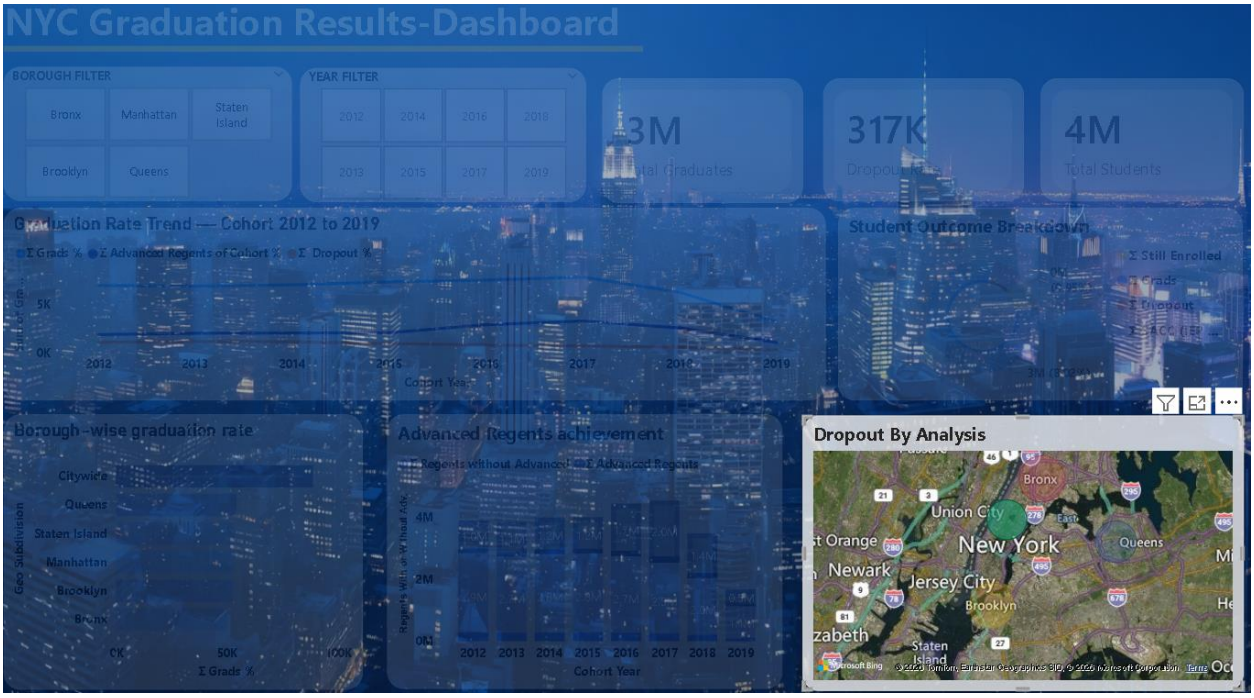
iii. Analysis Results

The geographic analysis shows that Bronx has a higher dropout rate compared to other boroughs, making it a high-risk area. In contrast, Queens and Staten Island show lower dropout rates, indicating better student retention. The use of bubble size and color helps in quickly identifying problem areas. This type of analysis is useful for policymakers to target specific regions for improvement.

iv. Visualisation

Visualisation Type: Bubble Map (Bing Maps integration in Power BI)

- Base map: Microsoft Bing Maps showing New York City and surrounding areas
- Bubble placement: Centred on each borough's geographic coordinates
- Bubble size: Proportional to Sum of Dropout count
- Colour gradient: Red (high risk) to Green (low risk) based on dropout severity


NYC Graduation Results-Dashboard
BOROUGH FILTER
Bronx
Manhattan
Staten Island
Brooklyn
Queens
YEAR FILTER
2012
2014
2016
2018
2013
2015
2017
2019
317K
4M
Total Students
Graduation Rate Trend — Cohort 2012 to 2019
Σ Grads %
Σ Advanced Regents of Cohort %
Σ Dropout %
Cohort Year
Student Outcome Breakdown
Σ Still Enrolled
Σ Grads
Σ Dropout
Borough-wise graduation rate
Citywide
Queens
Staten Island
Manhattan
Brooklyn
Bronx
Geo Subdivision
Σ Grads %
Advanced Regents achievement
Regents without Advanced
Σ Advanced Regents
Cohort Year
Dropout By Analysis
Bronx
Union City
New York
Queens
Newark
Jersey City
Brooklyn
Staten Island

Obj 5	**Advanced Regents Achievement Analysis** To examine the number of students who achieved Advanced Regents diplomas and analyze patterns of higher academic achievement across different cohorts and boroughs.

i. General Description

This analysis focuses on evaluating students who achieved Advanced Regents diplomas to understand academic excellence. It studies trends over time and compares high-level achievements among students.

ii. Specific Requirements

- Cohort Year
- Advanced Regents Students
- Regular Regents Students

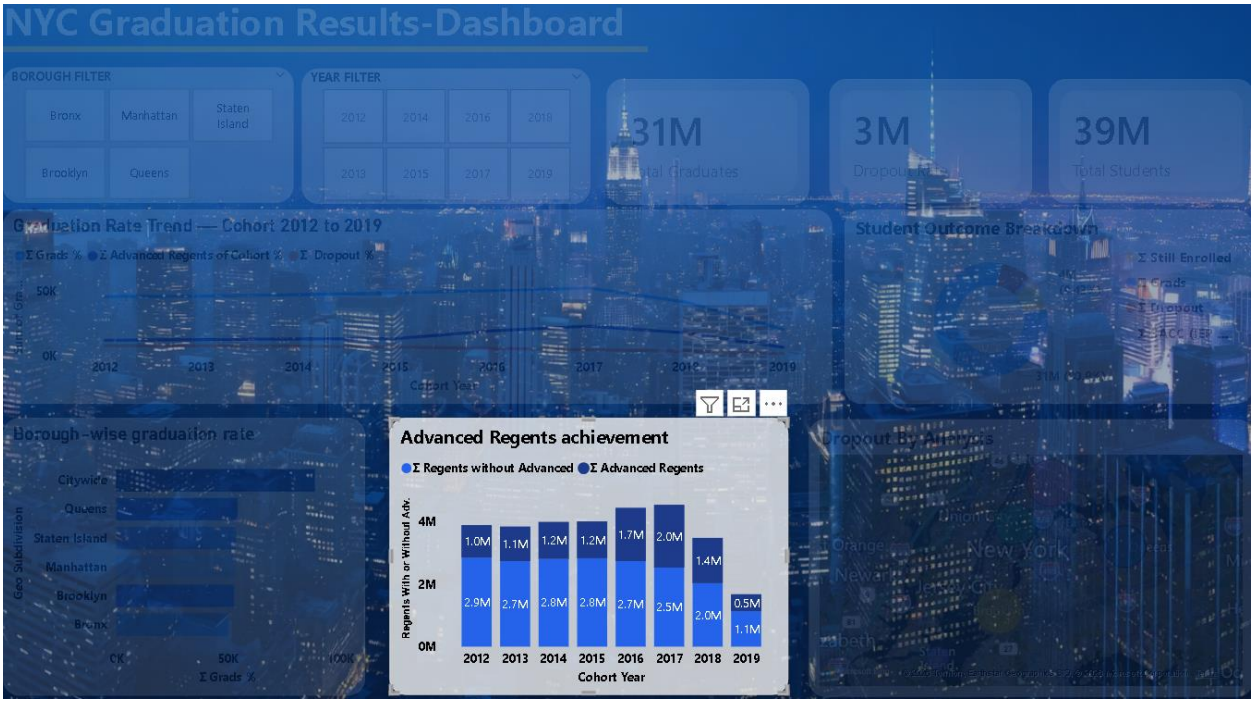
iii. Analysis Results

The analysis shows that the number of students achieving Advanced Regents diplomas increased steadily until 2017, indicating improvement in academic excellence. However, after 2017, there is a slight decline, suggesting inconsistency in high-level achievement. This highlights the need for continuous efforts to maintain academic standards.

iv. Visualisation

Visualisation Type: Stacked Bar Chart

- X-Axis: Cohort Year (2012 to 2019)
- Y-Axis: Regents With or Without Advanced (0M to 4M range)
- Two series: Light blue (Regents without Advanced) and Dark blue (Advanced Regents)
- Data labels showing exact figures (in millions) for each stack segment
- Chart titled "Advanced Regents achievement"


NYC Graduation Results-Dashboard
BOROUGH FILTER
Bronx
Manhattan
Staten Island
Brooklyn
Queens
YEAR FILTER
2012
2014
2016
2018
2013
2015
2017
2019
31M
3M
39M
Total Students
Student Outcome Breakdown
Borough-wise graduation rate
Citywide
Queens
Staten Island
Manhattan
Brooklyn
Bronx
Advanced Regents achievement
Σ Regents without Advanced
Σ Advanced Regents
Regents With or Without Adv.
4M
2M
0M
1.0M
1.1M
1.2M
1.2M
1.7M
2.0M
1.4M
0.5M
2.9M
2.7M
2.8M
2.8M
2.7M
2.5M
2.0M
1.1M
2012
2013
2014
2015
2016
2017
2018
2019
Cohort Year

5. Conclusion

This project successfully developed an interactive Power BI dashboard for analysing NYC high school graduation results across cohort years 2012 to 2019.The main aim of this study is to understand trends in graduation rates, dropout rates, and overall student outcomes across different boroughs. By using various visual tools such as charts, graphs, and maps, the data is presented in a clear and meaningful way.

- Overall Graduation Performance: NYC maintains a strong overall graduation rate of approximately 80.9%, representing 31 million graduates across the study period. This demonstrates the fundamental effectiveness of the city's educational system at scale.
- Borough Disparities Persist: Significant performance gaps exist between boroughs, with Queens and Staten Island consistently outperforming Bronx and Brooklyn. These disparities are likely linked to socioeconomic factors, school resource allocation, and community support structures.
- Dropout Risk is Geographically Concentrated: The bubble map analysis makes it visually evident that dropout risk is disproportionately concentrated in the Bronx. Targeted, borough-specific interventions are necessary rather than a one-size-fits-all approach.
- Student Outcomes Extend Beyond Graduation: The 9.42% still-enrolled population and the dropout segment indicate that a meaningful cohort of students require extended timelines or alternative educational pathways to achieve completion.
- Advanced Regents Achievement Peaked in 2017: The trend in Advanced Regents attainment shows growth up to 2017 followed by a decline, suggesting that while more students were pursuing higher academic credentials in the mid-period, recent cohorts may need additional academic support and encouragement.

In conclusion, this Power BI dashboard serves as a powerful analytical tool for NYC educational stakeholders, enabling evidence-based decision-making through clear, interactive, and actionable data visualisations.

6. Future Scope

This project can be further improved by incorporating more advanced techniques and additional data for deeper and more meaningful analysis. Including recent datasets will help in understanding current trends and making the analysis more relevant to present conditions. The use of advanced tools and technologies can significantly enhance the accuracy, efficiency, and usefulness of the results.

- School-level Granularity: Extending the analysis from borough-level to individual school-level data would enable identification of specific schools requiring intervention and those serving as models of excellence.
- Demographic Disaggregation: Incorporating student demographic variables such as race/ethnicity, gender, English Language Learner status, and disability classification would enable equity-focused analysis.
- Predictive Modelling: Applying machine learning techniques (logistic regression, decision trees) to predict dropout risk at the student level based on early academic indicators.
- Longitudinal College Enrolment Tracking: Linking graduation data with post-secondary enrolment information to measure the full educational pipeline outcome.
- Real-time Dashboard Updates: Connecting the Power BI dashboard to live NYC Open Data APIs to provide real-time updates as new cohort data becomes available.
- Comparative Analysis: Benchmarking NYC graduation performance against other major US cities (Los Angeles, Chicago, Houston) to contextualise findings within a national framework.
- Socioeconomic Overlay: Integrating poverty rate, median income, and unemployment data by borough to quantify the relationship between socioeconomic factors and educational outcomes.

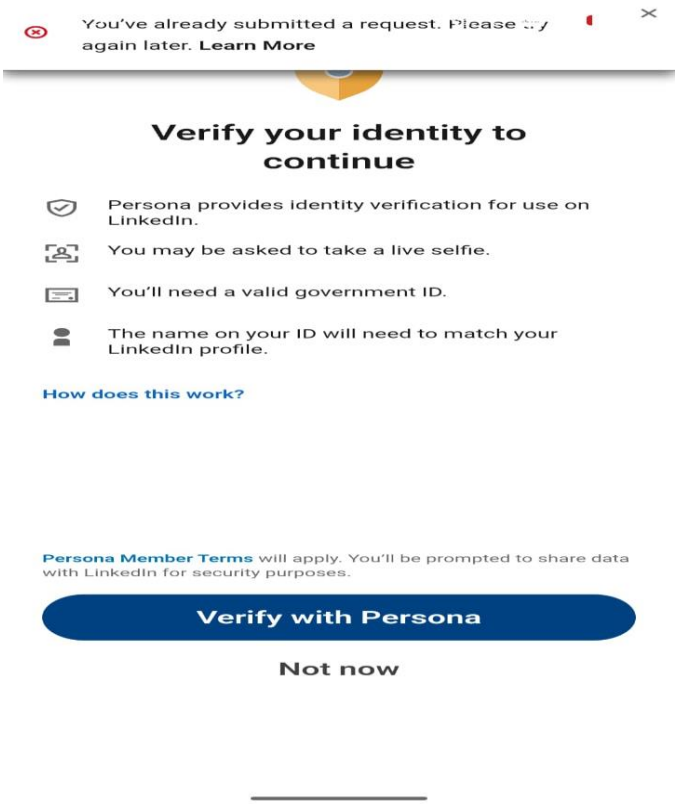
7. References

- New York City Department of Education. NYC Graduation Results Dataset. NYC Open Data Portal. https://catalog.data.gov/dataset/graduation-results-for-cohorts-2012-to-2019-classes-of-2016-to-2023
- Microsoft Power BI Documentation. Microsoft Corporation. https://www.microsoft.com/en-us/power-platform/products/power-bi/
- Power BI DAX Reference. Microsoft Docs. https://docs.microsoft.com/en-us/dax/

Link of the Project on LinkedIn

The Power BI dashboard and project documentation have been published on LinkedIn. Please visit the following link to view the interactive dashboard and project overview:

LinkedIn Project Link: LinkedIn is not working



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